

Jake Ward

github.com/jnward | jakenicholasward@gmail.com

Research Experience

MATS

January 2026–Present

Fellow [Alex Cloud & Alex Turner]

- Developing alignment methods which leverage weak oversight.

MATS

June 2025–December 2025

Scholar [Adam Shai & Paul Riechers]

- Novel methods for interpreting reasoning models using minimal parameter adapters.
- Studied how LLMs represent distributions over their beliefs about the world.

AISC

February 2025–April 2025

Mentee [Kola Ayonrinde (UK AISI)]

- Explored whether SAEs represent entity-attribute binding vectors.
- Examined dynamics of SAE feature splitting using semi-synthetic data.

University of Rhode Island

March 2019–July 2019

Research Assistant [Marco Alvarez]

- Developed and optimized ML pipelines for subject age estimation.

Publications

Satchel Grant, Victor Gillioz, **Jake Ward**, Thomas McGrath. “Shifting the Gradient: Understanding How Defensive Training Methods Protect Language Model Integrity.” *arXiv preprint arXiv:2604.16423*, 2026.

Nathan Hu, **Jake Ward**, Thomas Icard, Christopher Potts. “Transcoder Adapters for Reasoning-Model Diffing.” *arXiv preprint arXiv:2602.20904*, 2026.

Jake Ward, Paul Riechers, Adam Shai. “Rank-1 LoRAs Encode Interpretable Reasoning Signals.” *NeurIPS Workshop on Mechanistic Interpretability*, 2025.

Jake Ward. “Reasoning Elicitation is Scale-Dependent.” *NeurIPS Workshop on Efficient Reasoning*, 2025.

Jake Ward, Chuqiao Lin, Constantin Venhoff, Neel Nanda. “Reasoning-Finetuning Repurposes Latent Representations in Base Models.” *ICML Workshop on Actionable Interpretability*, 2025.

Industry Experience

Melange

February 2021–September 2023

Co-founder & Lead Engineer

- We raised a \$10M seed round to build an auction-based prediction markets platform.
- Later pivoted to building LLM agents for patent prosecution.
- Led entire engineering team of ~5 engineers.

Spark Insights

July 2019–May 2021

ML Engineer

- CV models to identify and predict damage to properties after natural disasters.

Education

University of Rhode Island

2015–2019

B.S. Computer Science